

JKBOSE 2026

CLASS 11TH

Morphology of Flowering Plants

LECTURE-05

BOTANY

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Topics To Covered

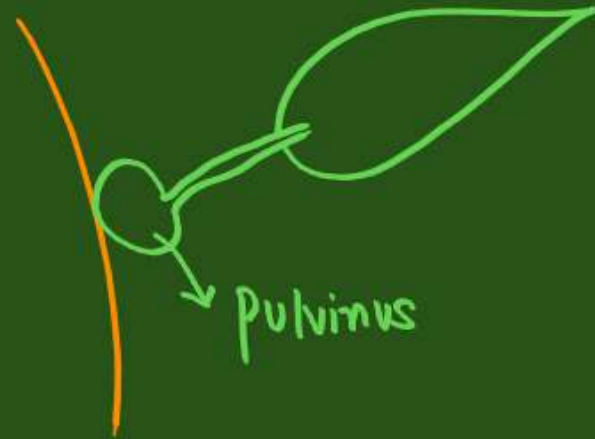
- 1** Modifications of leaf
- 2** Inflorescence
- 3** Structure of flower

MODIFICATIONS OF LEAF

① Modifications of leaf Base:

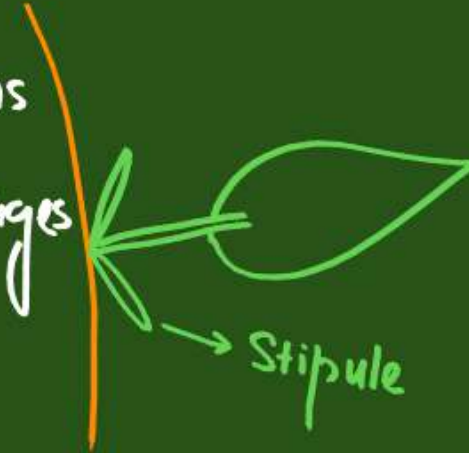
a. Pulvinus:

- Swollen leaf base.
- Ex. Pea.



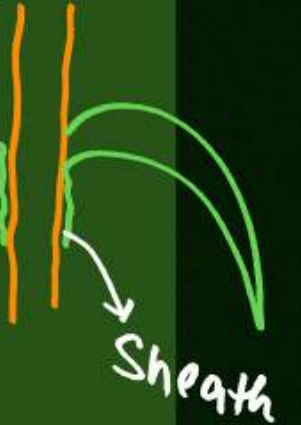
b. Stipule:

- leaf base forms leaf-like appendages kld as stipule
- Ex. Pea.



c. Leaf Sheath:

- leaf base forms sheath which partially or wholly covers stem.
- Ex. Monocots.



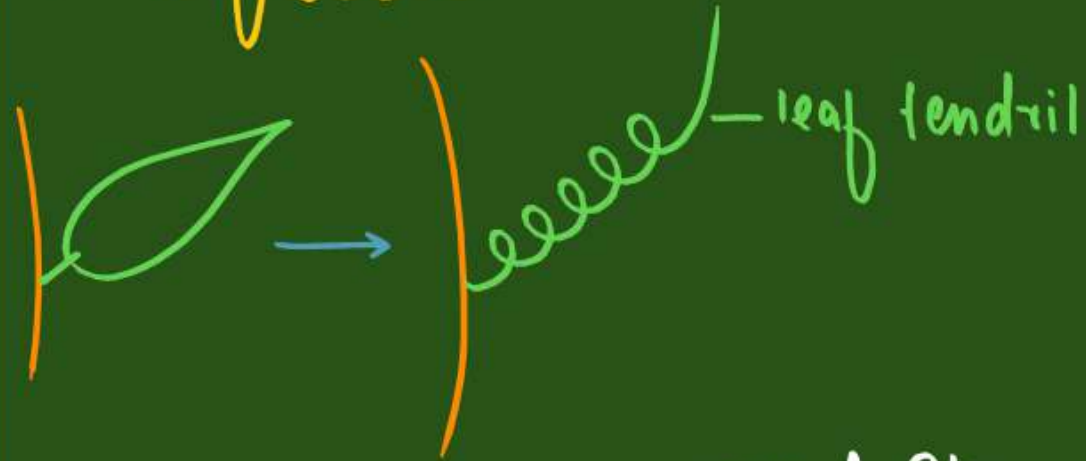
⑤ Modifications of Petiole :

a) Phyllode :

- It is xerophytic adaptation.
- In some xerophytic plants, petiole gets flattened and forms Phyllode which performs photosynthesis.
- Here in these plants — leaf-lets are short lived and fall early.
- Ex. Australian acacia.

③ Other modifications:

a. Leaf tendril:



- leaf forms spirally coiled str. kld as leaf tendril (helps in climbing)
- Ex. pea.

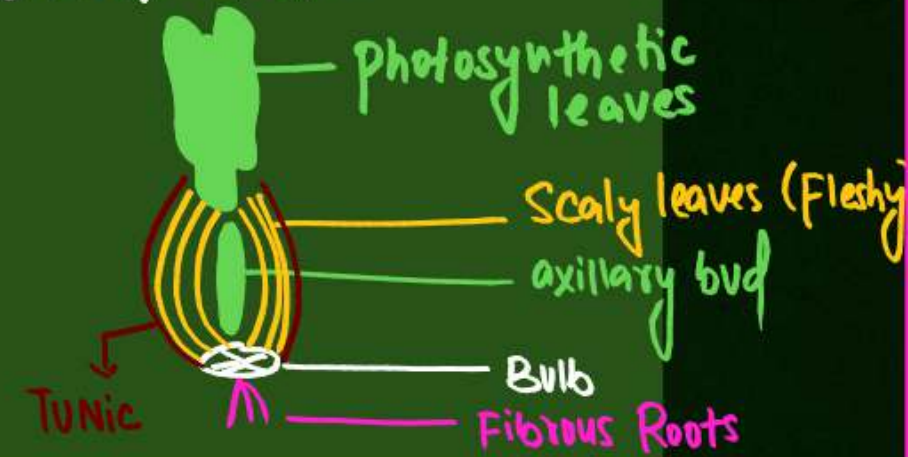
b. Spines:



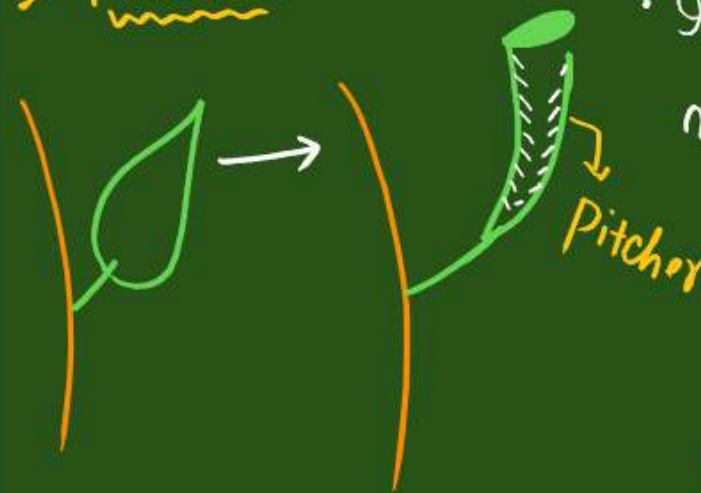
- In xerophytic plants, leaves form spines provides protection.
- Ex. Cactus.

c. Scaly leaves:

- For storage of food.
- Ex. Onion, Garlic

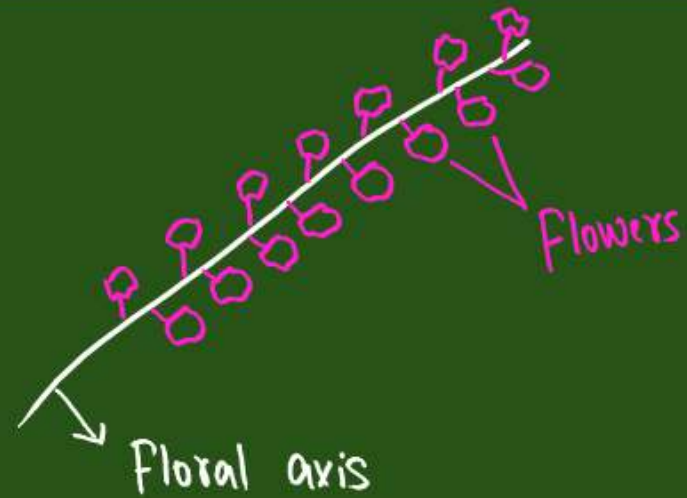


d. Pitcher:



- In insectivorous plants, leaf lamina modifies to form pitcher to trap insects.
- They eat insects to overcome N_2 -deficiency.
- Ex. Venus fly trap, Bladderwort, Nepenthes.

INFLORESCENCE



Branch/axis which bears flowers is kld as — Floral axis

or

Peduncle

or

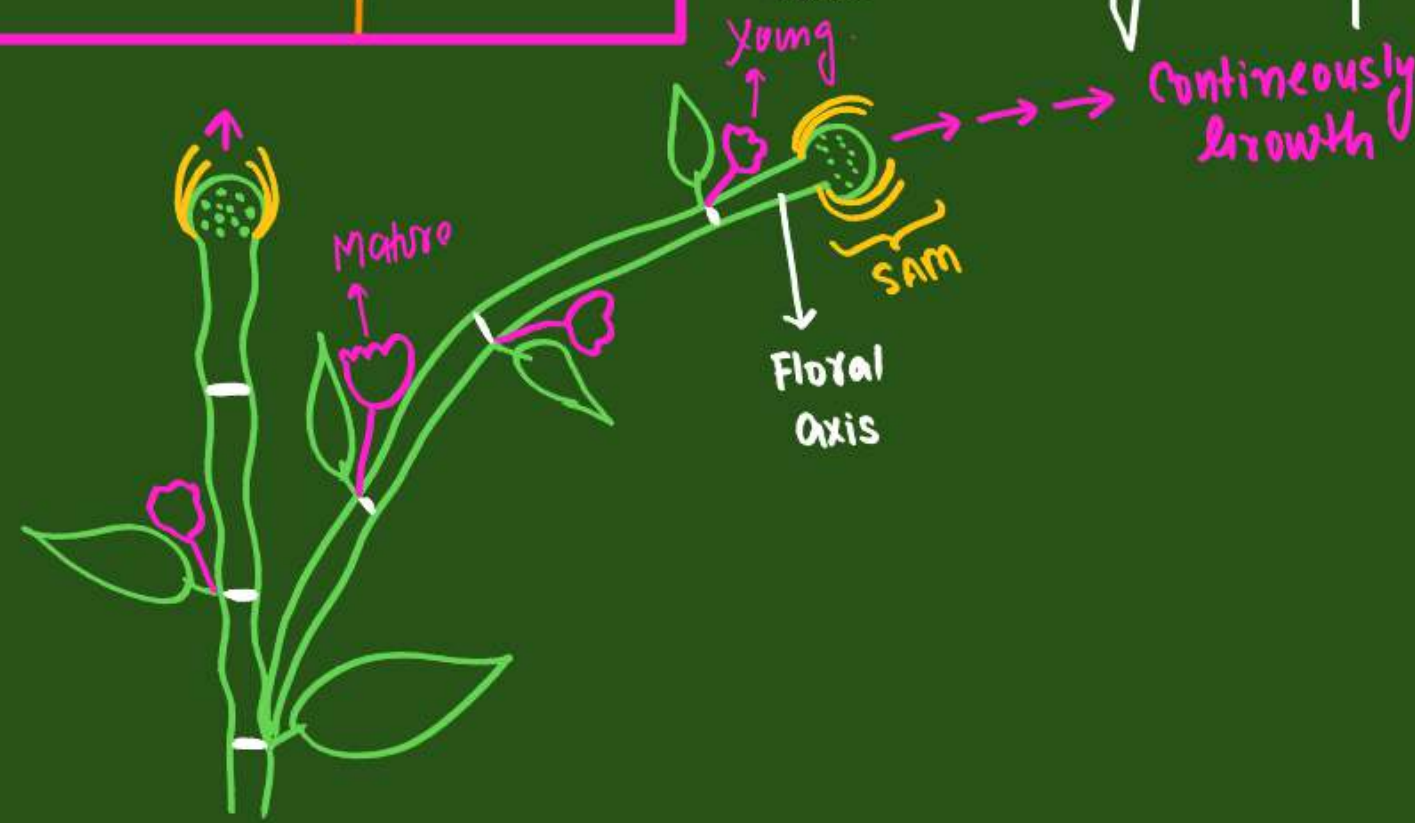
inflorescence axis.

- The pattern of arrangement of flowers on floral axis / peduncle / inflorescence axis is kld as **Inflorescence**.

- Inflorescence - 03 types:

- a) Racemose inflorescence.
- b) cymose inflorescence.
- c) Solitary inflorescence.

A) Racemose inflorescence



Note: Axillary bud forms Branch / flower.

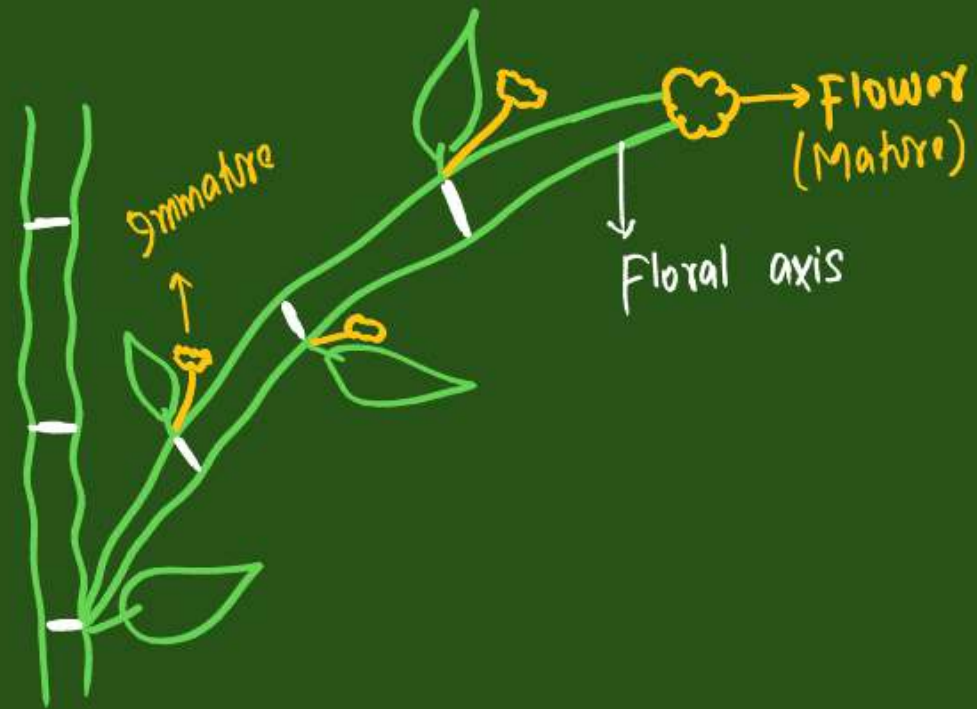
∴ In Racemose inflorescence, floral axis Shows Unlimited growth because of presence of SAM (Apical bud).

∴ Pattern of arrangement of flowers on floral axis is **Acropetal** → Young / Immature Flowers are towards Apex.

∥ Ex. Brassicaceae, Fabaceae.

Old / Mature flowers are towards base.

B) Cymose inflorescence:

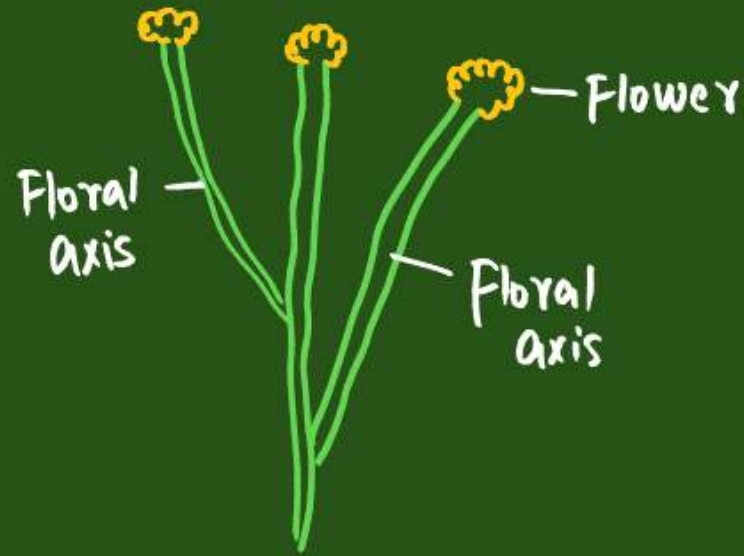


- Here - floral axis shows limited growth due to formation of flower from SAM (Apical bud)
- Pattern of arrangement of flowers is **Basipetal** → Mature / old flower at Apex.
↓
Young / Immature flowers towards base.
- Ex. Solonaceae,
Liliaceae.

c) Solitary Inflorescence:

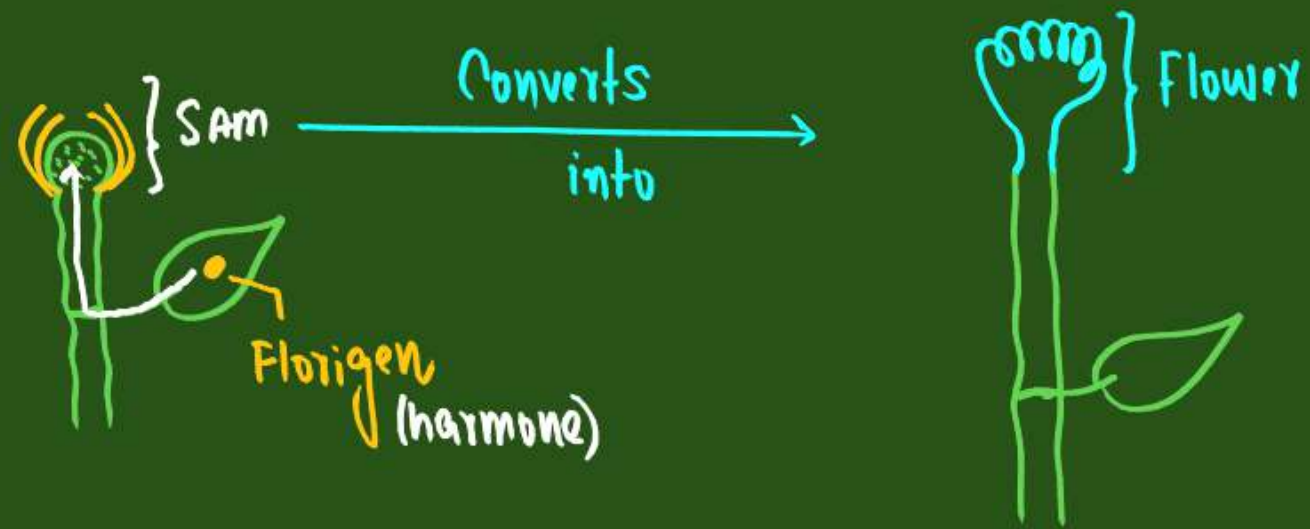
- Here - Single flower is present at tip of floral axis.

- ex. Poppy plant (Papaver somniferum)

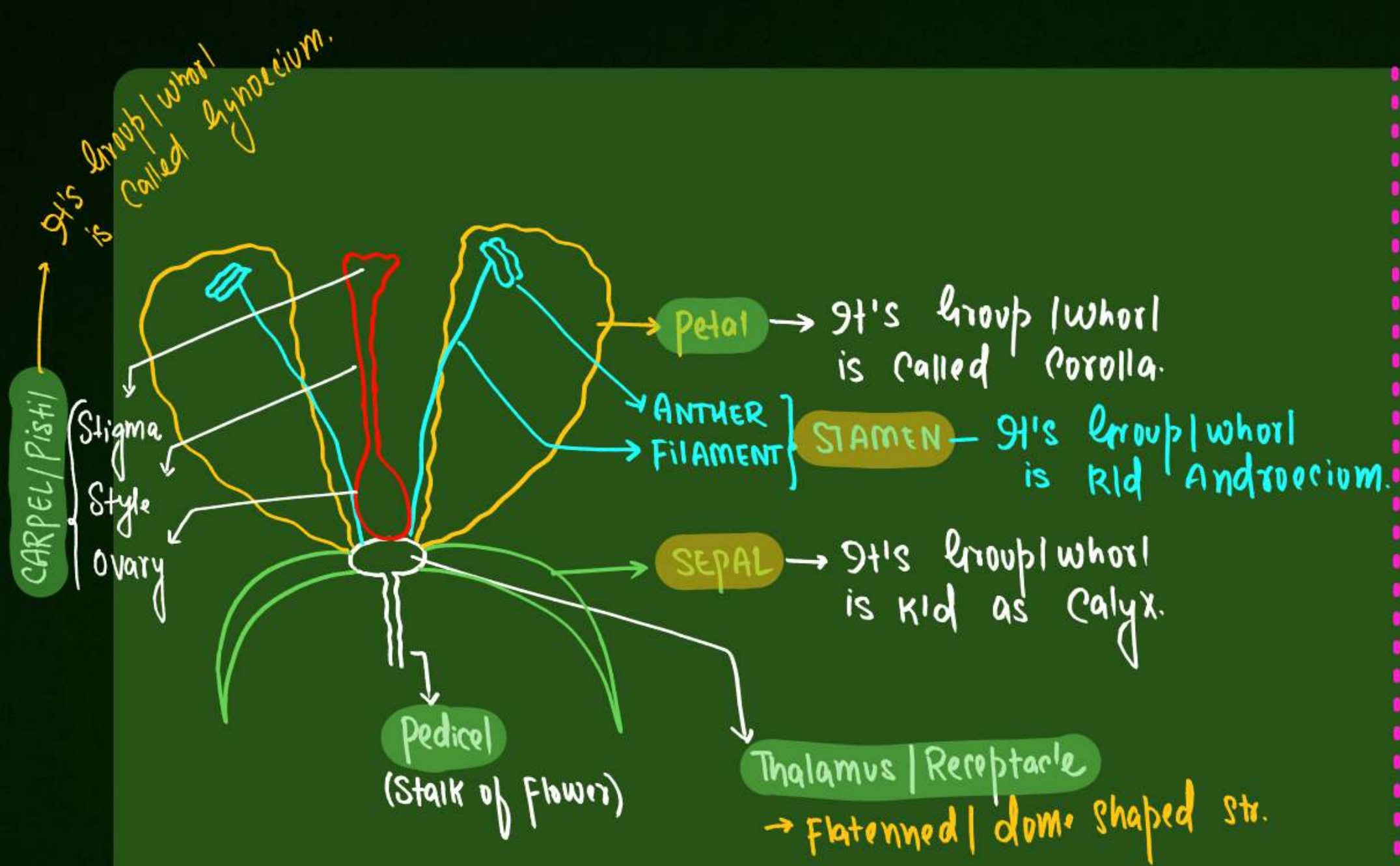


Flower STRUCTURE

- It is a modified and condensed shoot.



- ① Flower → possess modified leaves called as whorls.
- Internodes are Condensed
- It is meant for Sexual reproduction in Angiosperms.



④ 04 whorls of flower:

- Calyx
 - Corolla
 - Androecium
 - Gynoecium
- Non-essential whorls
(∵ Not involved in Sexual reproduction)
- Essential whorls
(∵ involved in Sexual reproduction)

FLOWER TYPES



1. Dichlamydeous

Ex: Brassicaceae, Solanaceae

- proper Sepals and petals present.
- Androecium & Gynoecium present.

2. Monochlamydeous

Ex: Lily

- Sepals and petals are not distinguishable clearly
- They are called as Tepals.
- Whorl of tepals is called as Perianth.
- Androecium & Gynoecium } +nt

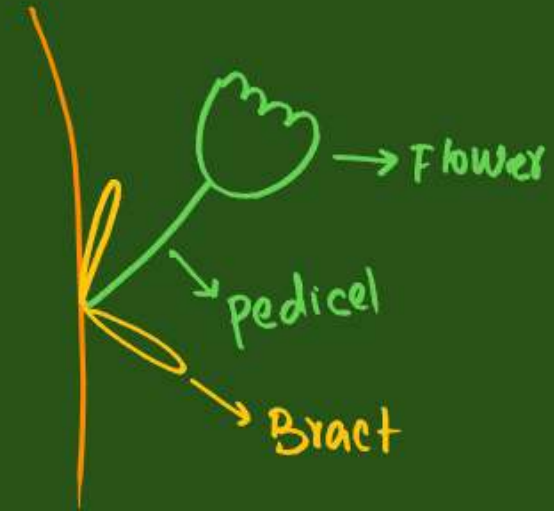
3. Achlamydeous

Ex: Euphorbia

- Sepals and petals Absent.
- Androecium and Gynoecium +nt.

FLOWER TYPES

1. Bracteate → Flower which possess **Bract**
→ represented by Br. leaf like appendages which arises from base of pedicel of flower



2. Ebracteate → Bract absent.
→ represented by Ebr.

FLOWER TYPES

1. Trimerous

Usually in monocots

• Floral appendages are in multiples of 03.

2 Tetramerous

Usually in dicots

• Floral appendages are in multiples of 04.

3. Pentamerous

Usually in dicots

• Floral appendages are in multiples of 05
• Most Common.

FLOWER TYPES (ON THE BASIS OF SYMMETRY)

ACTINOMORPHIC	ZYGOMORPHIC	ASYMMETRIC
Ex. Mustard, Datura, Chilli, Disc floret of sunflower	Ex. Pea, Bean, Cassia, Gulmohar Ray florets of sunflower 	Ex. Canna

FLOWER TYPES (ON THE BASIS OF OVARY POSITION)

HYPOGYNOUS	PERIGYNOUS	EPIGYNOUS
Ex. Mustard, China rose, Brinjal	Ex. Plum, Rose, Peach <i>Next lecture</i>	Ex. Guava, Cucumber, Ray florets of sunflower

QUESTIONS

Leaves become modified into spines in:

A Silk Cotton

B Opuntia

C Pea

D Onion

QUESTIONS

Leaves originate from and are arranged in an order.

- A** Root apical meristem, acropetal
- B** Floral meristem, basipetal
- C** Shoot apical meristem, acropetal
- D** Internode, basipetal

QUESTIONS

Swollen leaf base found in leguminous plants is called:

- A** Leaf blade
- B** Petiole
- C** Stipules
- D** Pulvinus

Thank
You